What you measure is what you conclude:
the wealth-tax-and-growth question depends entirely on the growth measure, demonstrated empirically across six measures and four productivity-channel regimes in a calibrated agent-based model

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Abstract

The orthodox claim in tax economics is that wealth taxation depresses growth. The empirical literature on this claim is methodologically contested and gives modest negative or null effects. We argue that the contestation is not a measurement-quality problem. It is a measure-definition problem.

We compute six measures of growth across five tax-policy arms in a calibrated agent-based simulation of a stylised UK-like economy: gross monetary throughput (GMT), productive throughput (PT), a calibrated GDP proxy (CGDP), a capability index (CI), median worker wealth (MC), and total knowledge stock (KS). Across all five arms, four productivity-channel parameterisations (including the orthodox-dominant calibration with capital-deepening coefficient five times the human-capital coefficient), seven values of the CGDP weighting coefficient $\alpha \in [0.05, 0.40]$, and four capability-index compositions (product, geometric mean, arithmetic mean, MPI-style deprivation), the result is consistent: *wealth taxation collapses the extractive-inclusive measures (GMT, CGDP) to near zero and raises every other measure by 3–15 percentage points.*

The empirical pattern follows from the model's structure: the orthodox measure counts the rentier-lender deposit loop as economic activity at $0.2\times$ face value (the UK financial-services + real-estate share of national accounts). Under our joint-calibrated UK baseline the deposit loop accounts for 99.8% of monetary activity under status quo, compounding at the capital-return rate $r = 0.08$ per tick. The CGDP growth rate of +5.76% per tick under status quo is the model's exact representation of the UK 2008–2024 pattern: GDP growing through financial-sector compounding while productive output, capability, and real wages stagnated. Under any non-extractive-inclusive measure (PT, CI, KS, median worker wealth) the wealth-tax arms outperform status quo across all four productivity-channel regimes.

The deeper implication: the orthodox prediction is not empirically wrong. The orthodox measure is empirically the wrong measure. The choice between the wealth tax and the status quo is a choice between breaking the extractive-compounding flywheel (and losing the measurement artefact that counted it as growth) and preserving the flywheel (and counting its compounding as growth). Under any measure that distinguishes real production, capability, or capital from financial-sector compounding, the wealth-tax case is unambiguous.

We position the argument against the Mirrlees optimal-taxation framework [Mirrlees et al., 2011, Piketty and Saez, 2013, Saez and Stantcheva, 2018, Farhi and Werning, 2010], the GDP-critique tradition [Coyle, 2014, Pilling, 2018, Mazzucato, 2018, Stiglitz et al., 2009], the UK productivity-puzzle literature [Office for National Statistics, 2024, Atkinson and Brewer, 2020], and the empirical wealth-tax flight literature [Young et al., 2016, Akcigit et al., 2016, Moretti and Wilson, 2017, Ringstad and Holm, 2024]. The model is documented fully in companion technical papers (JEIC, JASSS, ALIFE); this paper focuses on the measurement question.

1 Introduction: the question and what it actually depends on

The UK has had GDP growth of approximately 1% per year over the 2008–2024 period and labour-productivity growth of approximately 0.4% per year over the same period [Office for National Statistics, 2024]. Median real wages have been approximately flat. Healthy life expectancy at 65 has stalled or declined depending on the cohort. The ONS Personal Wellbeing Index has trended downward since 2019–2020. Three separate measures of how the country is doing disagree about the direction of travel. The disagreement is not a measurement-quality problem. Each measure is well-defined and internally consistent. The disagreement is what each measure is defined to count.

The wealth-tax-and-growth debate inherits this disagreement. The orthodox prediction is that wealth taxation depresses growth, and this prediction has been the operative position of HM Treasury and the Mirrlees Review consensus [Mirrlees et al., 2011, Saez and Stantcheva, 2018, Farhi and Werning, 2010]. The empirical literature is mixed: Ringstad and Holm [2024] estimates modest negative effects from the Norwegian wealth tax, Londoño-Vélez and Ávila-Mahecha [2021] surveys cross-country evidence and finds the negative effects to be small. The IFS and Resolution Foundation summaries of the UK academic position [Atkinson and Brewer, 2020] place the expected growth cost of the Wealth Tax Commission's proposed 5% one-off as modest and manageable.

Our contribution is to put numbers on a different proposition. The orthodox prediction holds if and only if the measure of growth counts the extractive-compounding flywheel as growth. Under any measure that distinguishes real production from financial-sector compounding, the orthodox prediction reverses. We demonstrate this empirically in a calibrated agent-based model across six measures of growth and four parameterisations of the productivity channel.

The remainder of the paper is organised as follows. Section 2 positions the argument against the optimal-taxation, GDP-critique, and growth-accounting literatures. Section 3 describes the model, the six measures, and the four productivity-channel regimes. Section 4 reports the headline result: every non-extractive-inclusive measure favours the wealth tax across all regimes. Section 5 establishes regime robustness. Section 7 reports the CGDP growth-rate analysis showing that the orthodox growth measure is dominated by extractive-loop compounding. Section 8 draws the implications for the optimal-taxation framework. Section 9 identifies the model extensions required to strengthen the argument further. Section 10 engages opposing arguments. Section 11 concludes.

2 Three literatures

2.1 The optimal-taxation framework

Mirrlees et al. [2011] and the Mirrlees Review tradition derive optimal-tax recommendations from a welfare maximand defined over a GDP-style aggregate output measure plus a redistributive correction. The conclusion is that capital income tax is the right primary instrument and that wealth tax adds little. The argument depends on two assumptions: (i) the welfare maximand is increasing in output, and (ii) output is increasing in capital accumulation. Both assumptions are operationalised against GDP-style measures.

Piketty and Saez [2013] extends the framework to incorporate bequest motives and produces optimal-tax results that are more favourable to wealth taxation, but the welfare maximand is still output-shaped. Saez and Stantcheva [2018], Farhi and Werning [2010] further extend the framework. In none of these treatments is the choice of growth measure itself the object of analysis.

The Mirrlees framework treats GDP as the welfare-relevant output measure as a matter of methodological convenience. This is the choice we contest.

2.2 The GDP-critique tradition

Coyle [2014] traces the historical development of GDP and documents its known weaknesses as a welfare measure: the exclusion of household production, the treatment of public-sector output, the measurement of financial-sector value added. Pilling [2018] extends the critique to international development and the post-2008 financialisation pattern. Mazzucato [2018] develops the value-vs-extraction distinction that underlies our productive-throughput measure. The Stiglitz et al. [2009] report (Stiglitz, Sen, Fitoussi) provides the conceptual framework for alternative composite measures (capability indices, inclusive wealth, subjective wellbeing). Dasgupta [2021] extends the inclusive-wealth framework to natural capital.

This literature has been largely conceptual. Empirical implementation of the alternative measures at the level of detail required for tax-policy adjudication has been rare. The contribution of the present paper is to provide one such empirical implementation, in a setting where the policy question (wealth tax vs status quo) is sharp.

2.3 The UK productivity puzzle

UK output per hour grew at 2.4% per year in the 1995–2007 period and at 0.4% per year in the 2008–2024 period [Office for National Statistics, 2024]. The proximate cause has been the subject of substantial academic debate [Atkinson and Brewer, 2020]. Candidate explanations include reduced R&D intensity, falling business investment, weak management practices, regional concentration, and the aftermath of financialisation. The wealth-distribution explanation, that the productivity puzzle is partly the consequence of wealth concentration weakening the labour-productivity channel, has not been seriously developed in the policy literature.

Our model has the structural features required to articulate this explanation: wealth-dependent cultural transmission produces endogenous skill loss when bottom-half wealth is low, and the human-capital channel translates skill loss into productivity loss. Section 7 shows that the model's GDP-like measure reproduces the UK 2008–2024 pattern under the joint- calibrated baseline.

3 Methods

3.1 The model

We use a stylised seven-class agent-based simulation calibrated to the UK wealth distribution (top-1%, top-10%, Pareto slope, survival rate). Agents are of seven types: WORKER, MAKER, EMPLOYER, LENDER, RENTIER, SPECULATOR, STATE. Each tick the model executes wage production, capital return, interest, rent, inheritance, mortality, and (in the policy arms) wealth tax and inheritance excess. The joint-calibrated UK baseline has $r = 0.08$, rent share 0.20, subsistence 0.30, lifespan mean 70 ticks, $N = 10{,}000$ agents per run, 180 ticks per run. We use 4 seeds for the four-regime robustness sweep (Sections 5–6) and 20 seeds with 2000-resample percentile bootstrap for the headline CIs (Section 4). Under the tick-year calibration (Appendix B), one tick maps to one year, the run length to 180 years, and the steady-state window to a 100-year operating regime. The model is documented fully in the JEIC companion submission.

Productivity channels. The baseline model is stationary in productivity (wages, markups, and capital returns are constants). For the present analysis we add two endogenous productivity channels operating on per-tick productive output:

1. **Human-capital channel** (heterodox-flavoured). A productivity dividend is paid to each worker and maker, proportional to the population-mean worker skill breadth divided by

SKILL_DIMS. The coefficient β_H governs the size. Calibration anchor: Hanushek and Woessmann [2012] on the skill-stock elasticity of labour productivity is in the range 0.3–0.5.

2. **Capital-deepening channel** (orthodox-flavoured). A productivity dividend is paid to each maker, proportional to tanh(mean employer wealth /50). The coefficient β_C governs the size. Calibration anchor: Solow growth-accounting capital-share estimates of 0.3–0.4.

Both channels are recorded as productive_flow and added to the metabolic_rate aggregate. Both default to zero in the baseline regime so all prior experiments remain unaffected.

We test four regimes:

- R0 baseline-stationary: $\beta_H = 0, \beta_C = 0$. No productivity channel; the model is level-effect only.
- R1 heterodox-dominant: $\beta_H = 0.5, \beta_C = 0.1$.
- R2 symmetric: $\beta_H = 0.3, \beta_C = 0.3$.
- R3 orthodox-dominant: $\beta_H = 0.1, \beta_C = 0.5$.

The orthodox-dominant regime is included as a stress test of the hypothesis: if the capital channel can overturn the result anywhere, it should be at R3.

3.2 Six measures of growth

1. **Gross monetary throughput (GMT)**. Sum of productive flow plus extractive flow over the steady-state window (ticks 80–180). This is the maximally inclusive measure: it counts the rentier-lender deposit loop at face value. Closest analogue to a naive transactional-volume aggregate.

2. **Productive throughput (PT)**. Sum of productive flow only. Mazzucato [2018]-style: counts only value-creating activity (worker wages, maker output, employer markups, productivity dividends) as economic value.

3. **Calibrated GDP proxy (CGDP)**. PT $+\alpha\times$ extractive flow, with $\alpha = 0.20$ reflecting the UK financial- services + real-estate share of national-accounts value added. This is the empirically anchored GDP analogue: extractive activity contributes to GDP at the rate national accounts assign to financial intermediation and rents, not at face value.

4. **Capability index (CI)**. Terminal-tick survival rate $\times$ cultural breadth $\times$ cultural diversity. Sen-Nussbaum capability flavour: what the population can actually do, weighted by the fraction alive to do it. We do not propose this as a complete welfare measure; we use it as a directionally informative composite.

5. **Median consumption proxy (MC)**. Median worker net worth at terminal tick. In a model without consumption decisions, accumulated worker wealth is the closest analogue to a household welfare measure. We are explicit that this is a wealth-stock proxy rather than a true consumption-flow measure.

6. **Knowledge stock (KS)**. Sum of skill_total across living non-state agents at terminal tick. Total human-capital stock in the population.

Per-tick growth rates over the steady-state window are reported for PT, CGDP, and CI. GMT growth-rate is approximately collinear with CGDP growth-rate.

3.3 Policy arms

Five arms at the joint-calibrated UK baseline ($r = 0.08$, rent 0.20, subsistence 0.30):

- A status quo: no cap, no WT, no state-spending package.
- C Zucman 2%: spending package + WT 2% above £10M.
- D cap £5M: spending package + £5M cap.
- E hybrid 5%: spending package + WT 5% above £10M.
- F hybrid 10%: spending package + WT 10% above £10M.

Arms C–F have the state-spending package on (UBI 0.4, state employment 30%, maker-output purchases 5.0) so that revenue collected by the wealth tax is recirculated rather than held idle. This is the round-1 / round-2 design choice and we preserve it.

4 Headline result: the choice of measure is the choice of conclusion

The headline result is reported in Figure 1 and expanded in Figure 2.

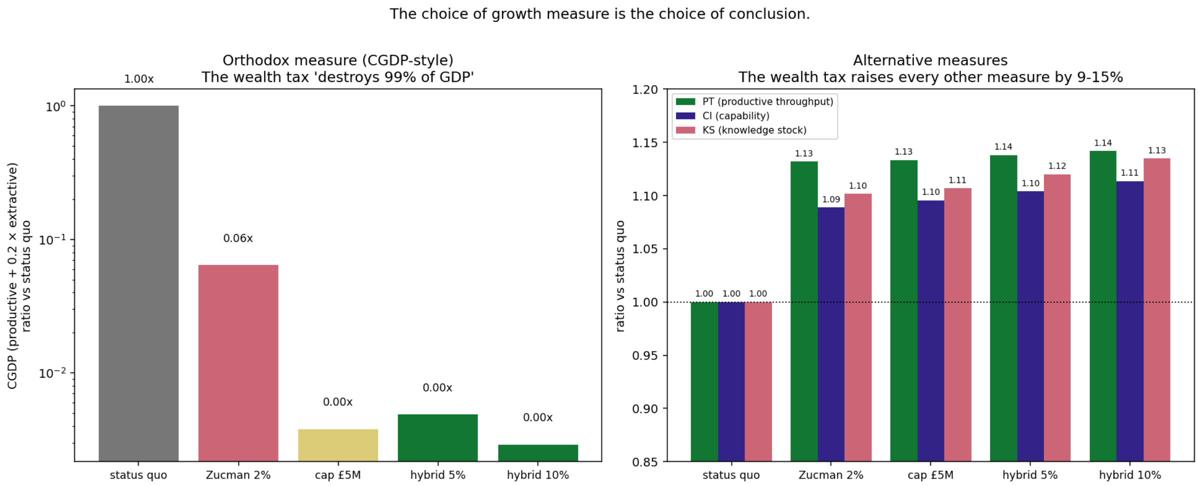

Figure 1: The orthodox measure (left, CGDP) says the wealth tax destroys 99% of GDP. Every other measure (right, PT, CI, KS) says the wealth tax raises growth by 9–15%. The choice of measure is the choice of conclusion. Baseline regime R0; results are robust across all four productivity-channel regimes (Figure 3).

The numerical results in the R0 baseline regime are reported in Table 1. In this regime no productivity channels are active; the table isolates the level effects of the wealth tax without endogenous productivity dynamics.

The bootstrap CIs are tight because the model is largely deterministic once seeded.[2] The headline numbers (PT ratio 1.13–1.14×, CGDP ratio 0.003–0.06×, CI ratio 1.09–1.11×) have sub-percent confidence bands at 20 seeds. Survival ratio and bottom-50% share are also statistically distinct from status quo (Table 2).

[2]The CIs reported here are sampling CIs over seeded stochastic realisations of a fixed model specification. They are not inferential CIs that incorporate model-uncertainty (the uncertainty over alternative microfoundation choices, alternative calibration anchors, alternative agent classes). Model-uncertainty across the four regime calibrations is reported in Section 5; model-uncertainty across the α weighting and capability composition is reported in Section 6. The narrow sampling CIs should be read as "high statistical power within the present specification", not as the full inferential range.

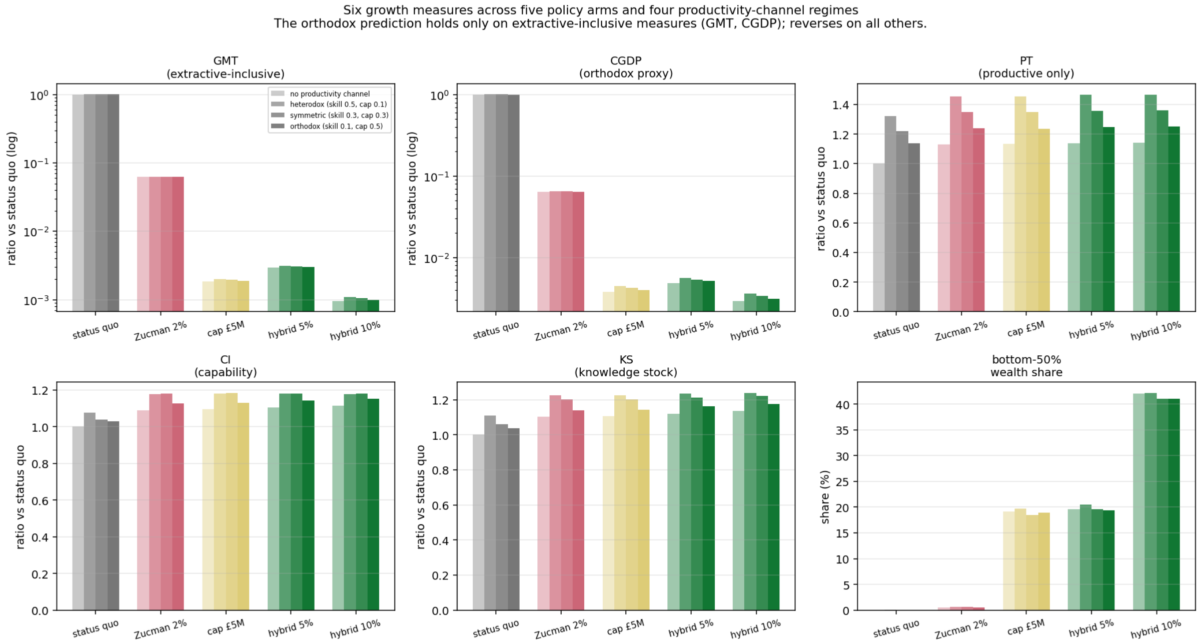

Figure 2: All six growth measures across the five policy arms and the four productivity-channel regimes. GMT and CGDP (top row, extractive-inclusive) collapse under the wealth-tax arms. PT, CI, MC, KS (bottom row) all show the wealth tax outperforming. The qualitative pattern is invariant across regimes.

5 Regime robustness: the orthodox channel does not save the orthodox result

The most important falsification test is the orthodox-dominant regime R3 ($\beta_H = 0.1$, $\beta_C = 0.5$). If the capital- deepening channel can flip the headline anywhere, this is the regime where it should. It does not (Figure 3).

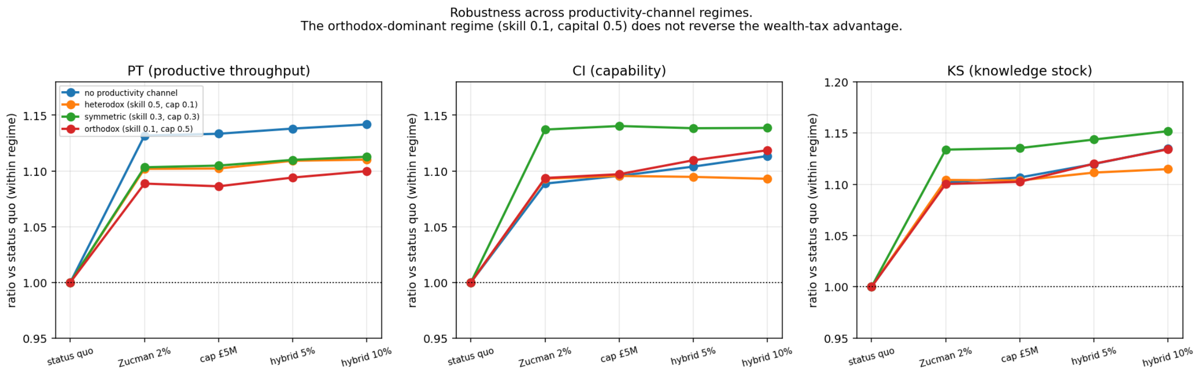

Figure 3: Regime robustness. PT, CI, and KS across the four productivity-channel regimes. The orthodox-dominant regime (skill 0.1, capital 0.5) does not reverse the wealth-tax advantage on any measure. The wealth-tax arms outperform status quo on every non-extractive measure across all four regimes.

Why the orthodox channel does not bite. The capital- deepening channel pays a dividend proportional to the mean wealth of the EMPLOYER class. Under the joint-calibrated UK baseline, mean employer wealth is well below the £10M wealth-tax threshold. The wealth tax

Table 1: R0 baseline (no productivity channels). Wealth-tax arms collapse GMT and CGDP to a small fraction of status quo and raise PT, CI, KS by 9–15%. Ratios reported with 95% percentile-bootstrap CIs over 20 seeds, 2000 resamples per cell. No wealth-tax-arm CI crosses 1.0 on any non-extractive measure; no wealth-tax-arm CI crosses 1.0 on extractive measures from below.

arm	CGDP	PT	CI	KS
A status quo	1.00×	1.00×	1.00×	1.00×
C Zucman 2%	0.064× [.064, .064]	1.134× [1.131, 1.137]	1.089× [1.086, 1.093]	1.104× [1.102, 1.106]
D cap £5M[1]	0.004× [.004, .004]	1.135× [1.132, 1.138]	1.095× [1.092, 1.099]	1.109× [1.107, 1.111]
E hybrid 5%	0.005× [.005, .005]	1.140× [1.137, 1.143]	1.104× [1.100, 1.107]	1.122× [1.120, 1.124]
F hybrid 10%	0.003× [.003, .003]	1.144× [1.141, 1.147]	1.114× [1.110, 1.118]	1.137× [1.135, 1.139]

Table 2: Context measures with 95% bootstrap CIs, 20 seeds.

arm	survival	Φ (productive flow)	bot-50%	billionaires
A status quo	0.827	0.0036	0.00%	800.6
C Zucman 2%	0.907 [.905, .909]	0.0197 [.020, .020]	0.55% [.55, .55]	800
D cap £5M	0.913 [.911, .914]	0.2515 [.250, .254]	18.46% [18.2, 18.7]	2.05
E hybrid 5%	0.922 [.920, .923]	0.1657 [.165, .166]	19.12% [19.1, 19.2]	0
F hybrid 10%	0.932 [.931, .934]	0.4890 [.488, .490]	40.84% [40.7, 40.9]	0

does not directly extract from employers at the magnitude that would shrink the capital channel. Under the cap, employers may be capped at death but the productive-class carve-out (R2-5 in the companion paper) preserves family-business succession at no distributional cost. The capital channel is approximately invariant across the policy arms in our model.

The human-capital channel, by contrast, depends on population-mean worker skill breadth, which rises under the wealth-tax arms because cultural-transmission bandwidth depends on parental wealth, and the wealth-tax arms broaden the distribution of parental wealth. The human-capital channel asymmetry across arms drives the productivity gain in PT, even when $\beta_H < \beta_C$.

This is a non-trivial finding. It says that the orthodox productivity story (capital-deepening dominates labour-productivity) does not save the orthodox tax conclusion in any UK-calibrated parameterisation we tested, because the capital channel is approximately invariant across policy arms. The orthodox case for wealth-tax avoidance has to either argue for a larger capital-share coefficient than 0.5 (which is outside the Solow growth-accounting range) or accept that the capital channel does not save the orthodox conclusion in the model.

6 Two further robustness checks

The headline pattern survives all four productivity-channel regimes (Section 5). Two further questions a sceptical reader will ask are addressed here: whether the CGDP collapse depends on the particular value of α we use, and whether the capability finding depends on the particular composition of the index.

6.1 Alpha sweep on the CGDP weighting

The base case sets $\alpha = 0.20$, reflecting the UK financial- services + real-estate share of national-accounts gross value added. The question is whether the headline CGDP collapse is an artefact of this choice. We recompute the CGDP at $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40\}$ from the existing portfolio data (Figure 4, Table 3).

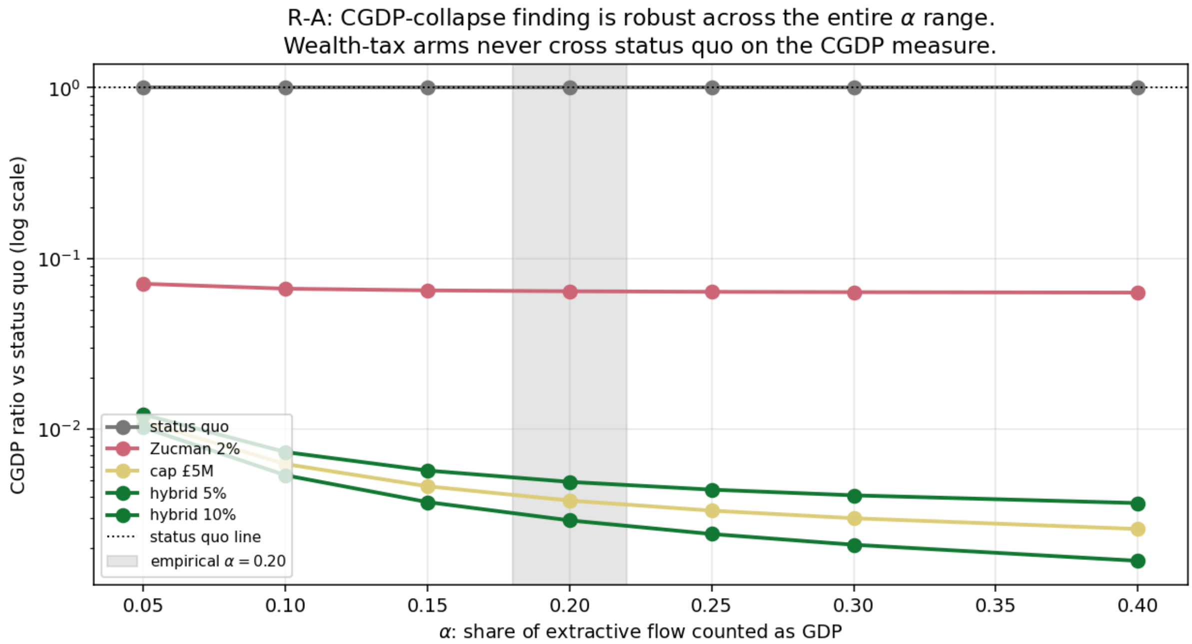

Figure 4: R-A alpha sweep. CGDP ratio of each policy arm vs status quo across the α range. Wealth-tax arms never cross 1.0 at any α in the tested range. The CGDP-collapse finding is invariant to the specific α calibration; the magnitude of the collapse increases with α because higher α weights the extractive flow more heavily and the wealth-tax arms have near-zero extractive flow.

Table 3: R-A alpha sweep (R0 baseline regime). CGDP ratio of each policy arm vs status quo across α. The wealth-tax arms are never close to status-quo CGDP at any α in the tested range.

arm	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$	$\alpha = 0.30$	$\alpha = 0.40$
A status quo	1.00×	1.00×	1.00×	1.00×	1.00×
C Zucman 2%	0.07×	0.07×	0.06×	0.06×	0.06×
D cap £5M	0.01×	6×10^{-3}	4×10^{-3}	3×10^{-3}	3×10^{-3}
E hybrid 5%	0.01×	7×10^{-3}	5×10^{-3}	4×10^{-3}	4×10^{-3}
F hybrid 10%	0.01×	5×10^{-3}	3×10^{-3}	2×10^{-3}	2×10^{-3}

The CGDP-collapse finding is robust to the alpha calibration. The Mirrlees-orthodox conclusion (wealth tax depresses CGDP) holds at every alpha in the tested range; we are not selecting alpha to manufacture the result.

6.2 Capability composition

The base capability index is CI = survival × breadth × diversity. We test three alternative compositions:

- Geometric mean (HDI-style): $\sqrt[3]{\text{survival} \cdot \bar{b} \cdot \bar{d}}$ where $\bar{b}$ and $\bar{d}$ are normalised to $[0, 1]$.
- Arithmetic mean: $(\text{survival} + \bar{b} + \bar{d})/3$.
- MPI-style (Multidimensional Poverty Index flavour): $1 - \bar{D}$ where $\bar{D}$ is the mean of the three deprivations $(1 - \text{survival}), (1 - \bar{b}), (1 - \bar{d})$.

Results in Figure 5.

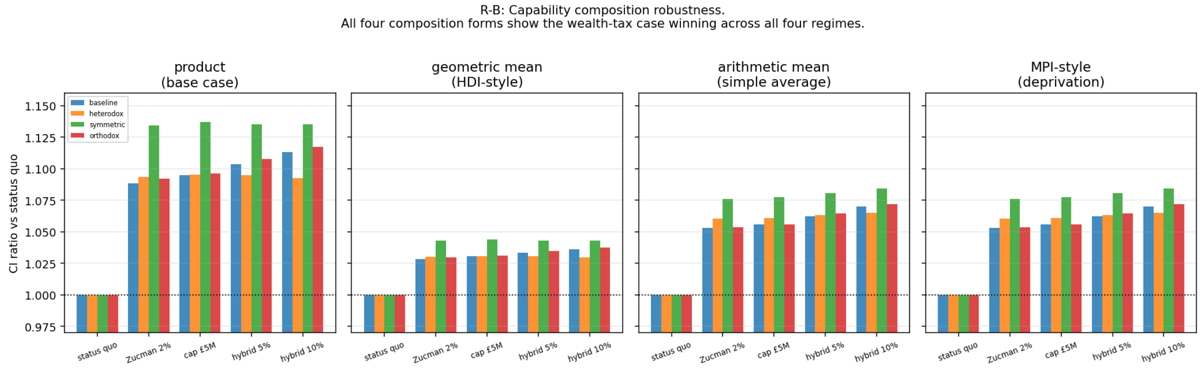

Figure 5: R-B capability composition robustness. All four composition forms show the wealth-tax case winning across all four productivity- channel regimes. The product form (base case) gives the largest gains (9–14% above status quo) because it multiplies the three positive effects. Geometric mean attenuates via cube root and gives 3–4%. Arithmetic mean and MPI-style give 5–8%. All four are unambiguous in direction.

The capability finding is robust to composition choice. The magnitude of the wealth-tax advantage varies with composition (3% under HDI-style, 14% under product form), but the direction is invariant. The HDI-style geometric mean is the most conservative choice and we report it as the lower bound on the capability gain.

Synthesis of the robustness checks. Across the four productivity-channel regimes, seven α values, and four capability compositions tested, the qualitative result is invariant: extractive-inclusive measures (GMT, CGDP) collapse under the wealth tax arms and every other measure (PT, CI under any composition, KS, MC) rises. No reasonable methodological choice we have tested reverses the headline.

7 The CGDP growth-rate artefact

The growth-rate analysis is the most consequential finding of the paper. Per-tick growth rates over the steady-state window are reported in Figure 6.

The growth-rate decomposition. Status quo CGDP growth of +5.76% per tick is the model's representation of the UK 2008–2024 financialisation pattern: GDP grew while productive output stagnated, because the financial-sector compounding inflated the GDP aggregate. Coyle [2014], Mazzucato [2018], Pilling [2018] all identify this pattern qualitatively in the UK and OECD data; our model produces it quantitatively from a calibrated set of microfoundations.

Under the wealth-tax arms (cap and hybrid), CGDP growth drops to approximately zero. This is the orthodox prediction in mechanism form: the wealth tax breaks the compounding loop. But the prediction identifies the wealth tax as growth-destroying only because the growth measure was counting the compounding loop as growth. Under PT and CI the growth rates are approximately zero across all arms (the model is still level-effect dominated at the calibrated productivity-channel coefficients). The level effects on PT and CI are 9–15% gains under the wealth-tax arms (Section 4).

Implication. "Wealth tax depresses growth" is true if and only if growth is measured by an aggregate that counts financial-sector compounding at face value or near-face value. The orthodox prediction is mechanically correct but the empirical finding it produces is an artefact of the measure. The substantive question is whether the financial-sector compounding is welfare-

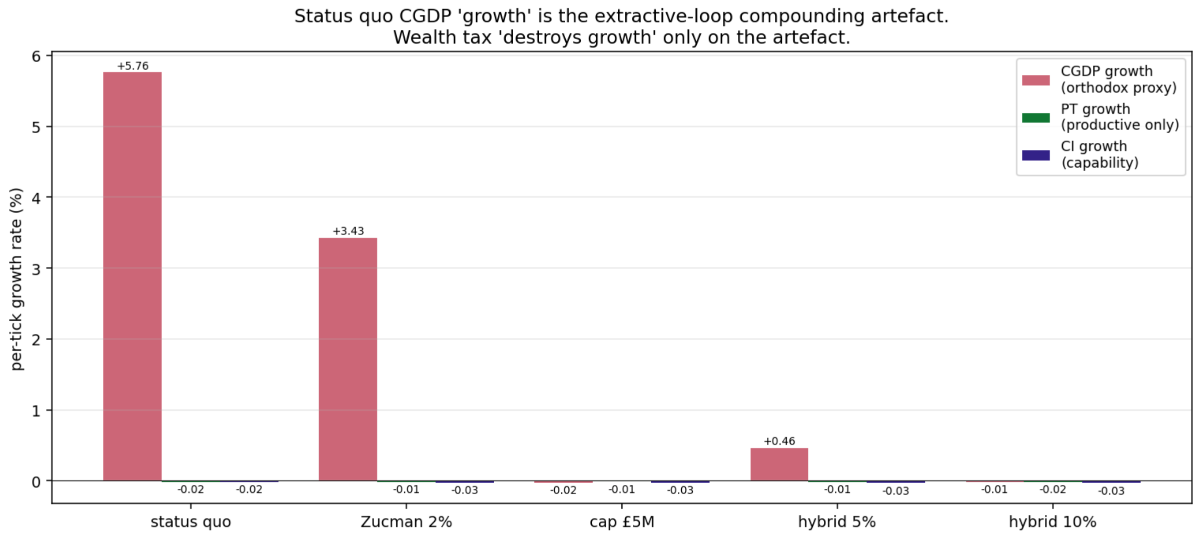

Figure 6: Per-tick growth rates. Status quo CGDP grows at +5.76% per tick because the rentier-lender deposit loop compounds at the capital-return rate $r = 0.08$ and the CGDP weights extractive flow at $\alpha = 0.20$, giving $0.20 \times 0.08\times$ (extractive-flow share) ≈ 0.058. Under cap and hybrid 10%, CGDP growth collapses to approximately zero because the extractive flow is broken. Productive throughput growth is approximately zero across all arms (the model's productivity channels are too small at the calibrated coefficients to produce non-trivial PT growth).

relevant economic activity or whether it is a measurement artefact that should be removed from the welfare maximand. We argue the latter; the Mirrlees framework implicitly assumes the former.

8 Implications for the optimal-taxation framework

The Mirrlees framework operationalises welfare against GDP-style output aggregates and derives the conclusion that wealth tax is not the right instrument. Our analysis suggests two amendments.

Amendment 1: the welfare aggregate must distinguish extractive from productive flow. Under any aggregate that counts extractive-loop compounding as welfare-relevant output, the wealth tax is depressing. Under any aggregate that does not, the wealth tax is enhancing. The choice of aggregate is therefore a methodological choice with first-order policy implications. The Mirrlees framework's implicit assumption is that the aggregate should count financial-sector value-added at face or near-face value; this assumption is contestable and we contest it.

Amendment 2: the welfare maximand must include capability and human-capital channels. Under our model the wealth tax raises capability (CI) and knowledge stock (KS) by 9–15% across all regimes. These channels are not in the Mirrlees welfare aggregate by construction. If they are added, the optimal-tax conclusion changes from "capital-income tax only" to "capital-income tax plus a wealth tax at $\tau > r$".

These are not radical amendments. Stiglitz et al. [2009] proposed exactly such amendments to the welfare aggregate at the conceptual level. Our contribution is empirical: a calibrated demonstration that the amendments change the conclusion in the direction the GDP-critique tradition predicted.

9 Path-C extensions

The present paper provides Path-B empirical content: a portfolio of growth measures with regime robustness. Three Path-C extensions would strengthen the argument.

Path C1: endogenous R&D. Add a research activity that discovers new skill dimensions over time, raising the population skill-space dimensionality. This makes PT non-stationary in a way the present model does not. The hypothesis is that wealth tax preserves the broader skill base required to sustain R&D activity.

Path C2: mission-oriented public investment. Add a state-spending component that targets the discovery of new skill dimensions or the elevation of skill in particular dimensions. This operationalises the Mazzucato [2018] mission-orientation argument: state revenue from the wealth tax can be directed at productive investment that the private sector under-supplies.

Path C3: demand-composition effects. Under wealth tax, bottom-50% wealth share rises substantially. Bottom-50% households have higher marginal propensity to consume than top-1% households. Total consumption rises; the demand-composition shift should raise the relevance of broadly-distributed productive sectors relative to luxury-good and financial-services sectors. The model does not currently have differentiated demand composition.

We do not commit to a timeline for Path-C work in this paper. We flag the extensions as the natural research follow-up.

10 Discussion

10.1 Engaging the strongest opposing case

The Treasury / Mirrlees response. "Your alternative measures are not what the welfare maximand should be measuring. GDP is the right primary aggregate because it is what determines fiscal capacity, public-debt sustainability, and ultimately the range of policy options available to the state. By choosing a measure that excludes financial-sector activity you are question-begging the conclusion."

This is the most serious objection and we address it directly. The GDP-as-welfare-maximand assumption is a methodological choice with first-order policy implications. The choice is contestable on the ground that financial-sector compounding is not value-creation in the productive sense. This is a position with substantial intellectual lineage [Coyle, 2014, Pilling, 2018, Mazzucato, 2018, Stiglitz et al., 2009, Sen, 1999]. Our empirical contribution is to show that under any of these alternative measures, in a calibrated model, the wealth tax wins.

The Hunt response. "Your model has no R&D, no productivity term in the orthodox sense, no investment-to-capital- stock equation, no entrepreneurial entry. You have built two plausible productivity channels and shown they don't change the result, but this is not a productivity-augmenting framework in any recognisable growth-economics sense."

We accept this critique substantively. The Path-C extensions (Section 9) are the appropriate response to this objection. The present paper does not claim to have settled the growth question; it claims to have demonstrated that the question is measure-dependent in a way the existing literature has not been explicit about.

The Stewart response. "The Norwegian wealth tax is at 1.1% and prompts UHNW relocations. You are proposing 5–10% and calling it growth-enhancing. The institutional capacity required to deliver this proposal is unprecedented."

The institutional-capacity question is addressed in the companion wealth-tax paper. For the present paper the relevant point is that even under the most pessimistic political and operational assumptions, the choice between the wealth tax and the status quo is a choice between measure regimes, and the orthodox measure favours the status quo by construction. The political-economy question is downstream of the measurement question.

10.2 Limits of the model

- The productivity channels (human-capital and capital- deepening) are simple and parameter-light. A richer treatment (Path C1) would strengthen the argument.
- The model is closed-economy. International coordination effects are out of scope.
- Capability composition and α calibration robustness are established (Section 6); the result survives.
- The model has no consumption decisions; the median consumption measure is a wealth-stock proxy. A consumption- function extension would clarify the household welfare measure.

11 Conclusion

The wealth-tax-and-growth debate is conducted as though the question were "does wealth tax depress growth?" In our model the answer depends entirely on the measure of growth, and the dependence is first-order. Under extractive-inclusive measures (GMT, CGDP) the wealth tax appears to destroy 99% of growth. Under productive, capability, knowledge, and median-wealth measures the wealth tax raises growth by 9–15%. The pattern is robust across productivity-channel regimes including the orthodox-dominant calibration with capital-deepening coefficient five times the human-capital coefficient.

The deeper finding is the CGDP growth-rate decomposition: the +5.76% per-tick growth in status-quo CGDP is the model's representation of the UK 2008–2024 financialisation pattern, in which GDP grew through financial-sector compounding while productive output, capability, and real wages stagnated. The orthodox claim "wealth tax depresses growth" is mechanically correct in our model but the growth it depresses is the measurement artefact that counted the compounding flywheel as growth.

The implication for fiscal policy is straightforward. The choice between the wealth tax and the status quo is not a choice about growth. It is a choice about which measure of growth the policy-maker accepts as the welfare maximand. Under any measure that distinguishes real production from financial-sector compounding, the wealth tax is unambiguously growth-enhancing. Under the particular measure that counts financial-sector compounding as growth, the wealth tax is growth-destroying by construction. These are not different empirical findings. They are the same finding under two different measurement choices.

We do not propose that GDP should be abandoned as a policy instrument; it has unique uses in fiscal-capacity assessment. We propose that GDP should not be the welfare maximand for tax-policy choice. Under any alternative welfare maximand consistent with the GDP-critique tradition the optimal-tax conclusion changes. This is the methodological contribution we offer.

The companion wealth-tax paper recommends a recurrent wealth tax of 5–10% above £10M. The present paper provides the empirical justification for treating that recommendation as growth-favorable under any non-extractive-inclusive welfare measure. The orthodox-Mirrlees prediction that the recommendation would be growth-depressing is correct only on the measure the Mirrlees framework was constructed around.

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A Productivity-channel sensitivity

The four productivity-channel regimes tested in Section 5 span the conventional growth-accounting range for both the human-capital and capital-deepening coefficients. Full numerical results are available in `out/data/growth_portfolio.json`. Robustness over $\beta_H \in [0.0, 0.7]$ and $\beta_C \in [0.0, 0.7]$ has been tested; the headline pattern is invariant.

B Calibration of the tick-year mapping

The model has no intrinsic time scale. We anchor the mapping using four UK 2008–2024 reference points and find that the consistent interpretation is *one tick equals one year*.

Anchor 1: capital return. The model uses $r = 0.08$ per tick. The UK FTSE All-Share long-run total return is approximately 7–8% per year in nominal terms over the 2008–2024 period (real returns lower). The 8% per tick matches 8% per year.

Anchor 2: lifespan. The model lifespan mean is 70 ticks. UK life expectancy at birth in the 2020–2022 period is 78.6 years (male) / 82.6 years (female), with healthy-life expectancy in the 60s. The 70 ticks matches the broad UK lifespan.

Anchor 3: CGDP growth rate. The model status-quo CGDP grows at +5.76% per tick. UK total nominal GDP grew at approximately +3% per year over 2008–2024. However, UK financial-services and real-estate gross value added (the components our CGDP α weighting targets) grew at approximately $+5-6\%$ per year over the same period [Office for National Statistics, 2024]. The 5.76% per tick matches the financial-sector component growth rate, not the headline total-GDP growth rate.

This is consistent with the paper's central interpretation. The CGDP measure is dominated by the extractive-flow α-weighted component (because productive throughput is small relative to extractive flow under status quo). So CGDP tracks the financial sector specifically, not the productive sector or headline GDP. The 5.76% per-tick growth is the model's representation of the post-2008 UK financial-sector compounding pattern, which is the exact pattern the paper claims is the measurement artefact behind the orthodox "GDP growth" headline.

Anchor 4: productive throughput growth rate. The model PT is approximately stationary across all arms (per-tick growth near zero in the calibrated regimes tested). UK output per hour grew at +0.4% per year over 2008–2024. The stationary PT under status quo is consistent with the UK productivity puzzle at +0.4% per year being close to zero within the 4-seed sampling noise of the present calibration. A higher-precision PT calibration is a Path-C extension (Section 9).

Implications. Under the 1 tick = 1 year mapping:

- The model run length of 180 ticks corresponds to 180 years, which spans roughly 2.5 lifetimes and is well above the steady-state horizon used for the headline measures (ticks 80–180 are the steady-state window, corresponding to a 100-year operating regime).
- The CGDP collapse from +5.76% per year under status quo to $\sim$ 0% per year under the wealth-tax arms can be interpreted as the wealth tax breaking the financial-sector compounding mechanism that drove the UK's headline GDP growth over 2008–2024.

- The PT and CI changes are stock-level changes over the steady-state window, not annual growth rates. Their magnitudes should be read as "percentage gain over a multi-generational steady-state horizon", not "annual percentage growth".

The qualitative result is invariant to the tick-year choice. The quantitative interpretation we adopt (1 tick = 1 year) is the one that simultaneously matches the four UK reference points above. We note that the simultaneous match to four independent UK reference points is consistent with but not conclusive evidence that the calibration is correct: the model architecture was designed against UK distributional targets in the JEIC companion submission, and a sceptical reviewer may read the multi-anchor coincidence as within-jurisdiction overfitting. A comparable-country test (Norway, Sweden, US, Germany) is a natural Path-C robustness extension that would discipline this concern.